

EcoSpendAI: AI-Powered Sustainable Expense Tracker

IBM SkillsBuild Internship Project Report

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Abstract

EcoSpendAI is an AI-powered sustainable expense tracking application designed to help users manage their daily expenses while promoting environmentally responsible spending habits. The application is developed using Python, Flask, HTML, CSS, JavaScript, SQLite, and ReportLab. Unlike traditional expense trackers, EcoSpendAI not only records financial transactions but also estimates the carbon dioxide (CO₂) emissions associated with different spending categories.

The system provides an interactive dashboard where users can analyze their spending patterns, monitor estimated carbon emissions, and receive intelligent eco-friendly spending suggestions. It also generates downloadable PDF expense reports for maintaining financial records. By combining artificial intelligence with sustainability, the project encourages users to make informed purchasing decisions that benefit both their finances and the environment.

This project supports the United Nations Sustainable Development Goal 12 (Responsible Consumption and Production) by promoting awareness of sustainable consumption through technology.

1. Introduction

Managing personal finances has become increasingly important in today's digital world. Many individuals use expense tracking applications to monitor their daily spending and improve their financial planning. However, most existing applications focus only on recording expenses and fail to educate users about the environmental consequences of their purchasing behavior.

EcoSpendAI addresses this limitation by integrating expense management with sustainability awareness. The application enables users to record daily expenses, classify them into categories such as Food, Travel, Shopping, and Utilities, and estimate the carbon dioxide (CO₂) emissions generated by those activities. This helps users understand how everyday spending choices affect the environment.

The project also provides AI-powered eco-friendly recommendations that encourage responsible consumption. Interactive charts and downloadable PDF reports make financial analysis simple and accessible. EcoSpendAI demonstrates how artificial intelligence and web technologies can be applied to solve real-world problems related to both personal finance and environmental sustainability.

2. Problem Statement

Traditional expense management systems primarily focus on tracking income and expenditure. Although they help users understand their financial habits, they do not provide any information regarding the environmental impact of daily purchases. As a result, users remain unaware of how activities such as shopping, transportation, and food consumption contribute to carbon emissions.

There is a growing need for a smart application that combines financial management with sustainability. Such a system should not only record expenses but also educate users about environmentally responsible spending habits and encourage conscious decision-making.

3. Objectives

The main objectives of EcoSpendAI are:

- Develop a user-friendly expense tracking application.
- Categorize daily expenses efficiently.
- Estimate CO₂ emissions based on spending categories.
- Provide AI-powered eco-friendly spending tips.
- Generate downloadable PDF expense reports.
- Promote responsible consumption and support UN SDG 12.

4. Features

EcoSpendAI includes several intelligent features that improve both financial awareness and sustainability.

Daily Expense Management: Users can add, edit, and manage expenses with category and amount.

CO₂ Emission Analysis: Every expense category is assigned an estimated carbon emission value, helping users understand environmental impact. **AI Eco-Friendly Tips:** The application generates intelligent sustainability suggestions based on spending behavior.

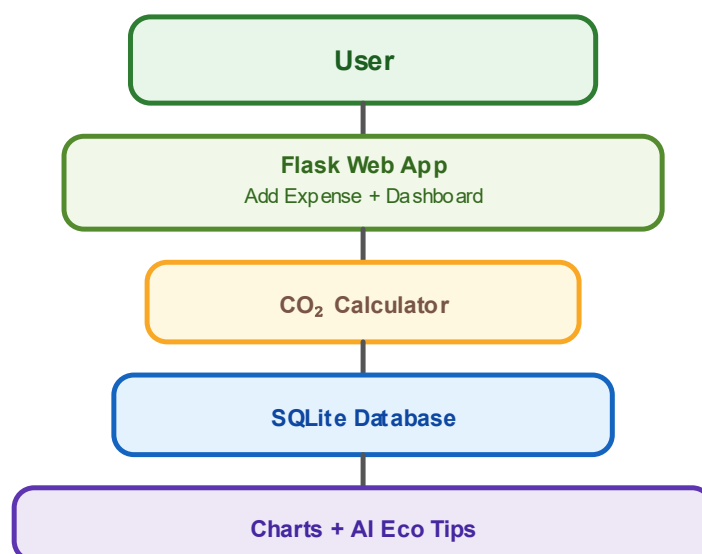
Interactive Dashboard: Users can view total spending, recent expenses, and spending summaries in one place.

PDF Report Generation: A professional expense report can be downloaded for future reference.

5. Technology Stack

Technology	Purpose
Python	Backend programming
Flask	Web framework
HTML	Page structure
CSS	User interface design
JavaScript	Interactive dashboard
SQLite	Local database
ReportLab	PDF report generation

6. System Architecture



The architecture consists of a Flask backend connected to an SQLite database. User inputs are processed by the application, stored securely, analyzed for carbon emissions, and displayed through the dashboard. The same data is also used to generate downloadable PDF reports.

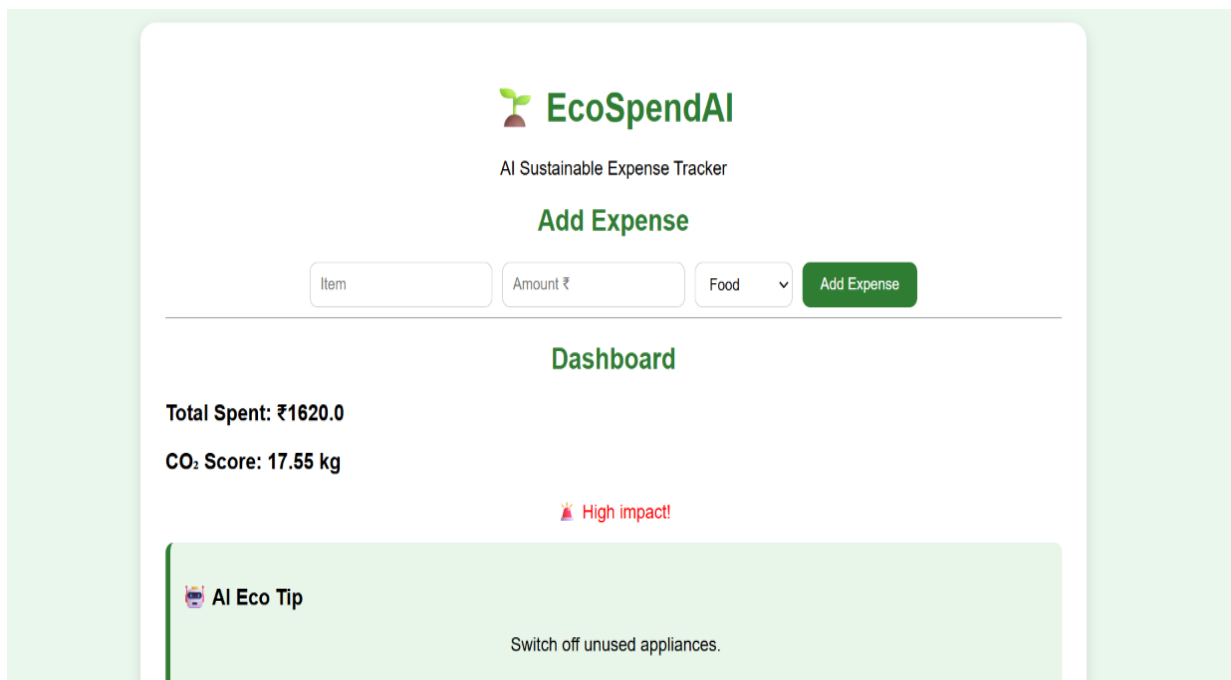
7. Working

1. User enters expense details.
2. Flask validates the input.
3. Data is stored in SQLite.
4. CO₂ emission is calculated.
5. Dashboard updates automatically.
6. AI displays eco-friendly recommendations.
7. User downloads the PDF report.

8. Results

The application was successfully developed and tested. Users can record expenses, analyze spending, estimate carbon emissions, and generate professional PDF reports. The dashboard provides an intuitive visualization of financial and environmental data, making the application suitable for personal finance management.

Home page



The screenshot displays the 'EcoSpendAI' application interface. At the top, the logo 'EcoSpendAI' is shown with a small green plant icon, followed by the subtitle 'AI Sustainable Expense Tracker'. Below this is a prominent green 'Add Expense' button. Underneath the button is a form with three input fields: 'Item', 'Amount ₹', and a dropdown menu currently set to 'Food'. To the right of these fields is a green 'Add Expense' button. A horizontal line separates the form from the 'Dashboard' section. The dashboard displays two key metrics: 'Total Spent: ₹1620.0' and 'CO₂ Score: 17.55 kg'. Below these metrics is a red warning icon and the text 'High impact!'. At the bottom, there is a green box containing an 'AI Eco Tip' icon and the text 'Switch off unused appliances.'

Dashboard

🔥 High impact!

🏠 AI Eco Tip

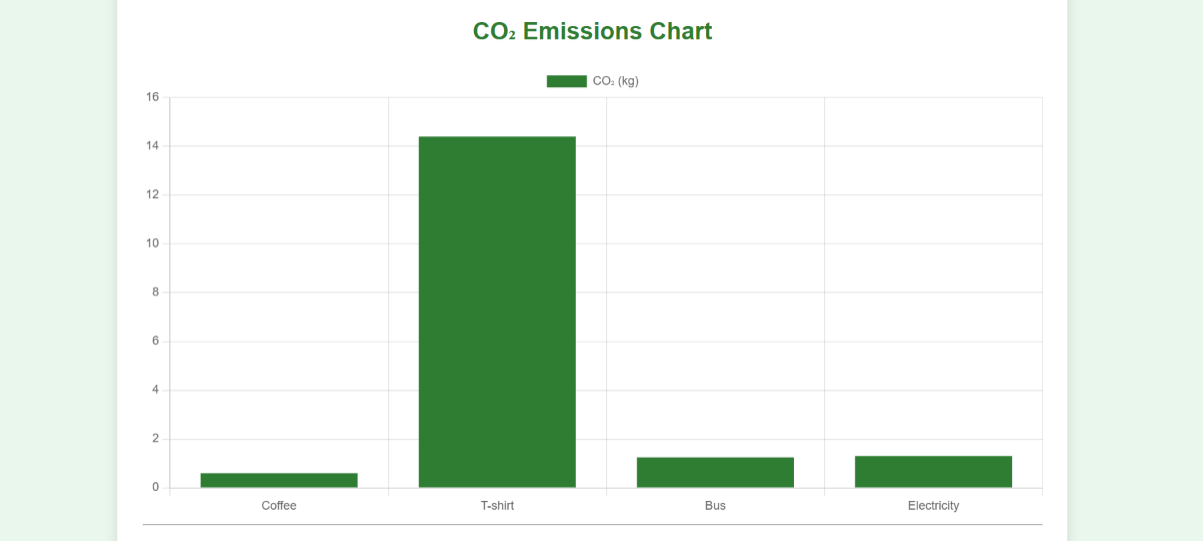
Switch off unused appliances.

📄 Export PDF

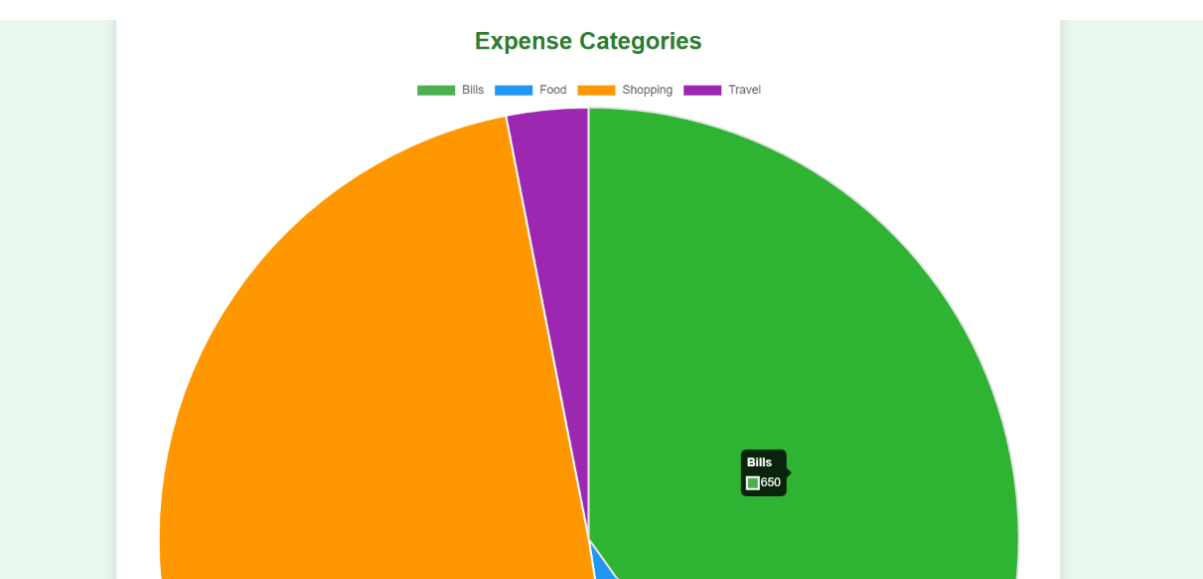
Recent Expenses

Item	₹	Category	CO ₂	Action
Electricity	650.0	Bills	1.3	🗑 Delete
Bus	50.0	Travel	1.25	🗑 Delete
T-shirt	800.0	Shopping	14.4	🗑 Delete
Coffee	120.0	Food	0.6	🗑 Delete

CO2 Emission Chart



Pie Chart



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Coffee | Rs.120.0 | Food | CO2 0.6 kg
T-shirt | Rs.800.0 | Shopping | CO2 14.4 kg
Bus | Rs.50.0 | Travel | CO2 1.25 kg

9. Conclusion

EcoSpendAI successfully integrates artificial intelligence with personal finance management to promote sustainable consumption. The application demonstrates how technology can help users become financially responsible while also reducing their environmental impact. Features such as CO₂ estimation, eco-friendly recommendations, interactive analytics, and PDF report generation make the system practical and user-friendly.

The project contributes directly to United Nations Sustainable Development Goal 12 by encouraging responsible consumption and production through intelligent decision-making.

10. References

1. Python Official Documentation
2. Flask Documentation
3. SQLite Documentation
4. ReportLab Documentation
5. United Nations Sustainable Development Goal 12

